

Mohammad Sabur Ali Mahdi

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EDUCATION

Queen Mary University of London

Integrated Master of Science (MSci) in Computer Science

London, UK

Sep. 2024 – June 2028

The Bridge Academy

A-Level: Computer Science A, Maths A*, Physics A*

London, UK

Sep. 2022 – June 2024

EXPERIENCE

EECS Lab Demonstrator

Queen Mary University of London

Oct. 2025 – Present

London, UK

- Guided 200+ first year Computer Science students through complex software engineering and OOP concepts, accelerating lab completion rates and improving program understanding
- Enhanced students' problem solving skills by conducting debugging sessions and teaching approaches to identify runtime errors, compiler issues and data structure flaws

AI Evaluator

Outlier AI

Jan. 2025 – Feb. 2026

Remote

- Evaluated 50+ AI generated code solutions for problem integrity and runtime complexity and ensured high accuracy across core software engineering tasks
- Designed edge case tests and prompts to detect and eliminate hallucinated logic, improving model reliability for generating code

Technology & Operations Work Experience

UBS

June 2023

London, UK

- Built a library tracking dashboard using Power BI, transforming unstructured records into structured tables and visual reports to improve visibility of library inventory

Virtual Technology Trainee

Springpod

Feb. 2023

Remote

- Explored core software engineering methodologies, gaining a foundational understanding of the full Software Development Lifecycle (SDLC), Agile sprint cycles, and system design pipelines

PROJECTS

NVIDIA Hack For Impact London | *React, NVIDIA Nemotron, RAPIDS, PostgreSQL* | [GitHub](#)

June 2026

- Built the shared PostgreSQL event store behind a five-agent city-operations platform, ingesting live London feeds (TfL, weather, air quality) with idempotent upserts to de-duplicate polled data
- Integrated NVIDIA Nemotron locally via an OpenAI endpoint, turning a RAPIDS (cuDF/cuGraph) relationship graph into grounded recommendations with all inference on device
- Integrated an ElevenLabs voice interface with a FastAPI and React admin panel to present real time impact reports while maintaining conversation memory across active user sessions

STEM Haqqathon | *RAG, Chrome Extensions API, Figma, React*

June 2026

- Designed a RAG based verification workflow to compare submitted social media content against trusted source records, improving response reliability by outputting with evidence rather than unsupported LLM generation
- Defined a confidence scoring and manual review process for classifying claims as verified, uncertain or requiring further scholar review, reducing the risk of low confidence content being presented as authoritative
- Prototyped the Chrome extension user journey in Figma and React, reducing friction by moving the verification flow from manual link submission towards in browser content checking for faster user feedback

LLM Dungeon Agent | *Python, Ollama, Local LLMs* | [GitHub](#)

June 2026

- Implemented response templates and validation layers in an offline Ollama environment, achieving 100% valid structured JSON outputs during final project testing
- Implemented action validation and logging to reject invalid model responses, improving reliability by tracking whether the agent could collect items, handle enemies, progress through levels, and reach objectives

- PetMinder App** | *React, Node.js, JavaScript, SQL, Figma, REST APIs* | GitHub Jan. 2026 – May 2026
- Developed core user features (calendars, messaging, profile management) using modular React components, resulting in a cohesive, smooth user experience
 - Connected React components to Node.js REST API endpoints and SQL data models, improving user specific data handling across booking records, pet minder profiles, emergency contacts, and messaging features
 - Traced booking, availability and file upload defects across React state, API responses and database records, validating fixes through 30+ manual functional test cycles
- Weather Navigation App** | *React, Google Maps API, OpenStreetMap API* | GitHub Jan. 2026 – Apr. 2026
- Built React settings workflows for driver preferences, notification controls, weather overlays, and help/FAQ pages, enabling users to customise route planning and weather risk alerts through a central interface
 - Integrated settings across planning, routing, and weather overlay modules by passing driver preferences, alert thresholds, and map coordinates into forecast and routing components, allowing driver specific weather risk calculations to update dynamically
 - Engineered a dynamic alerting system that processed multiple live forecast data (precipitation, visibility, wind) to trigger real time visual risk alerts and auto recalculate active routes
 - Resolved routing, destination validation, weather overlay, and driver risk issues by tracing incorrect API data handling, dropdown display conflicts, and asynchronous map update behaviour across React components and map services
- NHS Waiting List Database System** | *SQL, ER Modelling, Relational Design* | GitHub Nov. 2025
- Designed and normalised a relational database schema for patients, doctors, hospitals, appointments, and waiting list records, improving data organisation by reducing duplication and enforcing clear entity relationships
 - Implemented primary key, foreign key, and validation constraints to protect data integrity, ensuring patient and appointment records remained consistent across the database
 - Wrote 15+ SQL queries using joins, subqueries and aggregate functions to analyse waiting times and patient flow across regions, enabling clearer reporting on healthcare service demand
- Cosmic Showdown NEA Project** | *Python, OOP, SDLC Agile* | GitHub Sep. 2023 – Dec. 2023
- Built a 2D combat game framework in Python using object-oriented design, separating player logic, enemy behaviour, animations, inventory data, and game mechanics into modular classes to improve maintainability
 - Implemented vector based enemy tracking to calculate movement towards player positions during the game loop, enabling responsive enemy behaviour without noticeably affecting runtime performance

EXTRACURRICULAR

- Buddy Scheme Mentor** Sep. 2025 – Sep. 2026
Queen Mary University of London London, UK
- Appointed as an official peer mentor to guide first year students through their transition into university, sharing study strategies and guiding them through the year
- Programming Mentor** Oct. 2023 – Dec. 2023
The Bridge Academy London, UK
- Mentored secondary school pupils on programming basics, explaining core variables, conditions and basic loop debugging tactics using everyday analogies

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, PHP, SQL, HTML, CSS
Frameworks: React, Node.js, FastAPI, Tailwind CSS
Developer Tools: Git, GitHub, VS Code, Visual Studio, IntelliJ
Databases & Analytics: PostgreSQL, Power BI, ER Modelling, Relational Database Design
AI, Libraries & APIs: RAG, Ollama, Local LLMs, NVIDIA Nemotron, ElevenLabs, RAPIDS, cuDF, cuGraph, Chrome Extensions API, Google Maps API, OpenStreetMap API